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Experiment-3 : Data Ingestion and Cleaning

Aim:- Data Wrangling with Pandas: Manipulate and transform data to suit the analysis need.

Q1. What is data wrangling?. Explain the difference between Data Wrangling and Data preprocessing.

ANS:- Data wrangling is the process of cleaning, transforming, and organizing raw data into a more useful and structured format for analysis. It involves tasks such as handling missing values, filtering data, and reshaping datasets. Data preprocessing, on the other hand, is a broader term that includes all steps taken before applying machine learning algorithms, including data wrangling, feature scaling, encoding categorical data, and splitting datasets. In simple terms, data wrangling mainly focuses on cleaning and transforming data, while data preprocessing includes additional steps required to prepare data specifically for modeling.

```
import pandas as pd
data = pd.DataFrame({
    'Name': ['Ali', 'Sara', None, 'John'],
    'Age': [20, None, 22, 21],
    'Marks': [85, 90, None, 88]
})
print("Original Data:")
print(data)
data['Age'].fillna(data['Age'].mean(), inplace=True)
data['Marks'].fillna(0, inplace=True)
data.dropna(subset=['Name'], inplace=True)
print("\nAfter Data Wrangling:")
print(data)
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
data[['Marks']] = scaler.fit_transform(data[['Marks']])
print("\nAfter Data Preprocessing (Scaling):")
print(data)
```

Original Data:

	Name	Age	Marks
0	Ali	20.0	85.0
1	Sara	NaN	90.0
2	None	22.0	NaN

```
3 John 21.0 88.0
```

After Data Wrangling:

```
   Name  Age  Marks
0  Ali  20.0  85.0
1  Sara 21.0  90.0
3  John 21.0  88.0
```

/tmp/ipykernel_5452/566247699.py:9: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series consisting of rows or columns. Please use inplace=True to modify the DataFrame in place.
The behavior will change in pandas 3.0. This inplace method will never work because it is deprecated.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(value, inplace=True)'.

```
data['Age'].fillna(data['Age'].mean(), inplace=True)
/tmp/ipykernel_5452/566247699.py:10: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series consisting of rows or columns. Please use inplace=True to modify the DataFrame in place.  
The behavior will change in pandas 3.0. This inplace method will never work because it is deprecated.
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method(value, inplace=True)'.

```
data['Marks'].fillna(0, inplace=True)
```

After Data Preprocessing (Scaling):

```
   Name  Age  Marks
0  Ali  20.0   0.0
1  Sara 21.0   1.0
3  John 21.0   0.6
```

Q2. What is the importance of exploratory data analysis(EDA) in pandas?

ANS:- Exploratory Data Analysis (EDA) is the important step in Data analysis that helps in understanding the structure, pattern, and relationship with dataset. In Pandas, EDA allows user to summarize the data using the statistical measure, visualize trends, and detect anomalies and outliers. It helps in identifying missing value, understanding distribution, and making informed decisions about further data processing. EDA is essential because it provides insights into the data before applying any complex analysis or machine learning models.

```
import pandas as pd

data=pd.DataFrame({
    'Name': ['Ali', 'Sara', 'John', 'Aman'],
    'Age': [20, 21, 22, 23],
    'Marks': [85, 90, 88, 92]
})
print(data.info())
print(data.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0   Name     4 non-null        object
1   Age      4 non-null        int64
2   Marks    4 non-null        int64
dtypes: int64(2), object(1)
memory usage: 228.0+ bytes
None
```

	Age	Marks
count	4.000000	4.000000
mean	21.500000	88.750000
std	1.290994	2.986079
min	20.000000	85.000000
25%	20.750000	87.250000
50%	21.500000	89.000000
75%	22.250000	90.500000
max	23.000000	92.000000

Q3.What are the main differences between series, Dataframe,and Panel?

ANS:- In Pandas, Series, DataFrame, and Panel differ mainly in their structure and dimensionality:

Series: A one-dimensional labeled data structure. It contains a single list of values with an associated index. Similar to a single column in a table.

DataFrame: A two-dimensional labeled data structure. It consists of rows and columns, like a table or spreadsheet. Each column can store different types of data. It is essentially a collection of multiple Series.

Panel: A three-dimensional data structure. It was used to store multiple DataFrames together. It has three axes (items, major axis, minor axis). It is now deprecated and no longer commonly used.

Key Differences: Dimension: Series (1D), DataFrame (2D), Panel (3D) Structure: Series (single column), DataFrame (table), Panel (collection of tables) Usage: Series for simple data, DataFrame for tabular data, Panel for 3D data (obsolete)

```
import pandas as pd

#Series (1D)

s=pd.Series([10, 20, 30])

#DataFrame (2D)
```

```
df=pd.DataFrame({
    'A': [1, 2],
    'B': [3, 4]
})

print(s)

print(df)
```

```
0    10
1    20
2    30
dtype: int64
   A  B
0  1  3
1  2  4
```

Q4.Explain the use of:

a) isnull(): The isnull() function is used to check the missing or null value in a dataset. It returns a boolean result where the true indicates a missing value and indicates a non-missing value

```
import pandas as pd

df = pd.DataFrame({'A': [1, None, 3]})

print(df.isnull())
```

```
      A
0  False
1   True
2  False
```

b) notnull:- The notnull() function is used to check for non-missing values in a dataset. It returns the True for values that are present and false for missing values.

```
import pandas as pd
df=pd.DataFrame ({'A': [1, None, 3]})
print(df.notnull())
```

```
      A
0   True
1  False
2   True
```

c) `dropna()`:- the `dropna()` function is used to removed rows or coloums that contain missing values. By default, it remove rows with at least one null value.

```
import pandas as pd
df = pd.DataFrame({'A': [1, None, 3]})
print(df.dropna())
```

```
      A
0  1.0
2  3.0
```

d) `fillna()`:- The `fillna()` function is used to replace the missing values with a specified values such as zero, mean, or any constant.

```
import pandas as pd
df = pd.DataFrame({'A': [1, None, 3]})
print(df.fillna(0))
```

```
      A
0  1.0
1  0.0
2  3.0
```

Q5. what is column transformation and why is it important in data analysis?

ANS:- Column transformation refers to modifying or converting the values of the column to make them suitable for analysis. This may include data types, scaling values, applying mathematical operations, or creating new derived column. Column transformation is important because it improve the quality and usability of the data, making it easier to analyse and interpret. It also plays a key role in preparing data for machine learning models, where properly formatted data leads to better performance and accurate results.

```
import pandas as pd
df = pd.DataFrame({'A': [1, None, 3]})
print(df.isnull())
print(df.notnull())
print(df.dropna())
print(df.fillna(0))
```

```
      A
0  False
1   True
2  False
      A
0   True
1  False
```

```
2    True
    A
0    1.0
2    3.0
    A
0    1.0
1    0.0
2    3.0
```

Conclusion:- Data wrangling using Pandas is a crucial step in data analysis that helps convert raw data into a clean and structured format. Techniques like EDA, handling missing values, and column transformation improve data quality and make the analysis more accurate and reliable.